

Two Way ANOVA

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Load in packages

```
library(tidyverse)
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ ggplot2    3.5.1      ✓ tibble     3.2.1
## ✓ lubridate  1.9.3      ✓ tidyr      1.3.1
## ✓ purrr      1.0.2
## — Conflicts — tidyverse_conflicts() —
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(DescTools)
library(effectsize)
library(emmeans)
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```
library(car)
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:DescTools':
##
##      Recode
##
## The following object is masked from 'package:dplyr':
##
##      recode
##
## The following object is masked from 'package:purrr':
##
##      some
```

```
library(reghelper)
```

```
##
## Attaching package: 'reghelper'
##
## The following object is masked from 'package:DescTools':
##
##      ICC
##
## The following object is masked from 'package:base':
##
##      beta
```

Disable scientific notation

```
options(scipen = 999)
```

VARIABLES & STUDY PREMISE

- 120 participants, 40/group, between-subjects design
- DV - performance on final retrieval quiz
- VariableA (IV/PV) - level of trained expertise (categorical, binomial)
 - A1 - Novice (lower mean)
 - A2 - Expert (higher mean)
- VariableB (IV/PV) - study strategy intervention (categorical, binomial)
 - B1 - Restudy Intervention (lower mean)
 - B2 - Retrieval Intervention (higher mean)

Read in the data

```
final <- read.csv("finalexam-2way.csv")
head (final)
```

```
##      Subject VariableA VariableB    DV X      X.1      MEAN      SD X.2
## 1      S1      A1      B1 -1.03 NA      A1B1 -0.25000000 0.7882674 NA
## 2      S2      A1      B1 -1.93 NA      A1B2 -0.01066667 1.1183452 NA
## 3      S3      A1      B1  0.46 NA      A2B1 -0.58933333 0.8064479 NA
## 4      S4      A1      B1 -0.01 NA      A2B2  0.81400000 1.0435966 NA
## 5      S5      A1      B1  0.04 NA TOTAL TOTAL -0.00900000 1.0725759 NA
## 6      S6      A1      B1 -0.48 NA              NA          NA NA
##      X.3 MEAN.1 SD.1
## 1 A1 TOTAL      NA  NA
## 2 A2 TOTAL      NA  NA
## 3 B1 TOTAL      NA  NA
## 4 B2 TOTAL      NA  NA
## 5              NA  NA
## 6              NA  NA
```

Treat as factors:

```
final <- read.csv('finalexam-2way.csv', stringsAsFactors = TRUE)
head(final)
```

```
##      Subject VariableA VariableB    DV X      X.1      MEAN      SD X.2
## 1      S1      A1      B1 -1.03 NA      A1B1 -0.25000000 0.7882674 NA
## 2      S2      A1      B1 -1.93 NA      A1B2 -0.01066667 1.1183452 NA
## 3      S3      A1      B1  0.46 NA      A2B1 -0.58933333 0.8064479 NA
## 4      S4      A1      B1 -0.01 NA      A2B2  0.81400000 1.0435966 NA
## 5      S5      A1      B1  0.04 NA TOTAL TOTAL -0.00900000 1.0725759 NA
## 6      S6      A1      B1 -0.48 NA              NA          NA NA
##      X.3 MEAN.1 SD.1
## 1 A1 TOTAL      NA  NA
## 2 A2 TOTAL      NA  NA
## 3 B1 TOTAL      NA  NA
## 4 B2 TOTAL      NA  NA
## 5              NA  NA
## 6              NA  NA
```

One-way ANOVA VariableA

```
model_a <- lm(DV~ 1 + VariableA, data=final)
anova(model_a)
```

```
## Analysis of Variance Table
##
## Response: DV
##           Df  Sum Sq Mean Sq F value Pr(>F)
## VariableA   1    1.767   1.7666   1.5426 0.2167
## Residuals 118 135.133   1.1452
```

No, there is not significant difference between groups. Same performance regardless of assigned expertise.

One-way ANOVA VariableB

```
model_b <- lm(DV~ 1 + VariableB, data=final)
anova(model_b)
```

```
## Analysis of Variance Table
##
## Response: DV
##           Df  Sum Sq Mean Sq F value      Pr(>F)
## VariableB   1   20.238  20.2377    20.47 0.00001454 ***
## Residuals 118 116.662   0.9887
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Yes, there is a significant difference between groups. Different performance regarding study strategy intervention.

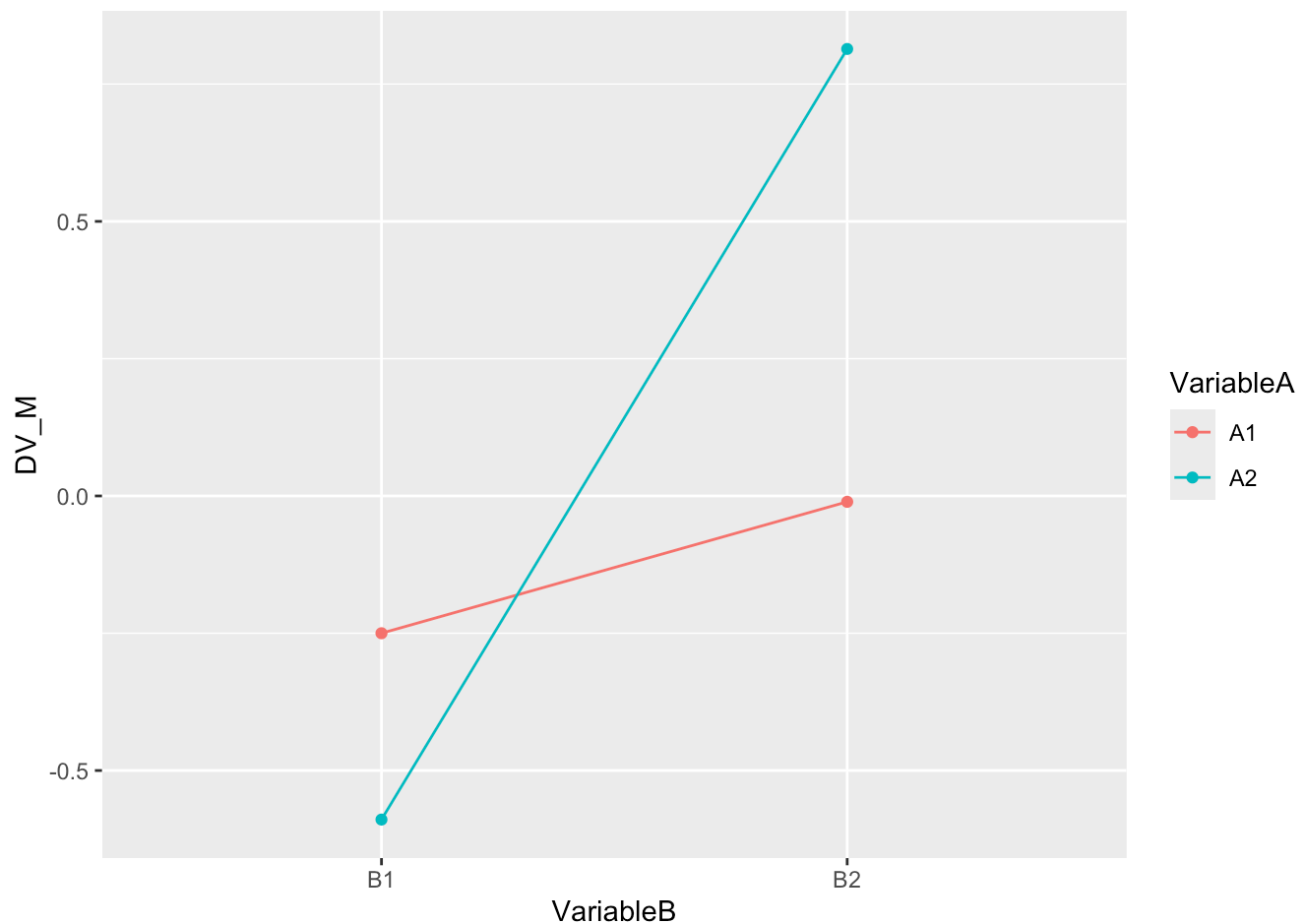
```
model_1 <- lm(DV ~ VariableA*VariableB , data=final)
summary(model_1)
```

```
##
## Call:
## lm(formula = DV ~ VariableA * VariableB, data = final)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.41933 -0.55150  0.07433  0.63300  1.86600
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -0.2500     0.1735  -1.441  0.15226
## VariableAA2     -0.3393     0.2453  -1.383  0.16929
## VariableBB2      0.2393     0.2453   0.976  0.33133
## VariableAA2:VariableBB2  1.1640     0.3470   3.355  0.00107 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9502 on 116 degrees of freedom
## Multiple R-squared:  0.235, Adjusted R-squared:  0.2152
## F-statistic: 11.88 on 3 and 116 DF, p-value: 0.0000007721
```

Visualizing the data

```
final %>% group_by(VariableA, VariableB) %>%
  summarize(DV_M=mean(DV)) %>%
  ggplot(aes(x=VariableB, y=DV_M, color=VariableA, group=VariableA)) +
  geom_point() + geom_line()
```

```
## `summarise()` has grouped output by 'VariableA'. You can override using the
## `.groups` argument.
```



Performance on final quiz (z-scores): A2B2 best (Expert, Retrieval); A1B2 second (Novice, Retrieval); A1B1 third (Novice, Restudy); A2B1 last (Expert, Restudy).

Contrasts

```
contrasts(final$VariableA)
```

```
##      A2
## A1    0
## A2    1
```

```
contrasts(final$VariableA) <- cbind(c(.5, -.5))
colnames(contrasts(final$VariableA)) <- c('Restudy versus Retrieval on Expertise')
contrasts(final$VariableA)
```

```
##      Restudy versus Retrieval on Expertise
## A1                      0.5
## A2                     -0.5
```

```
model_x <- lm(DV~ 1 + VariableA, data=final)
summary(model_x)
```

```
##
## Call:
## lm(formula = DV ~ 1 + VariableA, data = final)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.44233 -0.68983  0.06033  0.69717  2.56767
##
## Coefficients:
##                                Estimate Std. Error t value
## (Intercept)                 -0.00900    0.09769  -0.092
## VariableARestudy versus Retrieval on Expertise -0.24267    0.19538  -1.242
##                                Pr(>|t|)
## (Intercept)                  0.927
## VariableARestudy versus Retrieval on Expertise  0.217
##
## Residual standard error: 1.07 on 118 degrees of freedom
## Multiple R-squared:  0.0129, Adjusted R-squared:  0.004539
## F-statistic: 1.543 on 1 and 118 DF,  p-value: 0.2167
```

```
contrasts(final$VariableB)
```

```
##      B2
## B1    0
## B2    1
```

```
contrasts(final$VariableB) <- cbind(c(.5, -.5))
colnames(contrasts(final$VariableB)) <- c('Novice versus Expert on Study Strategy')
contrasts(final$VariableB)
```

```
##      Novice versus Expert on Study Strategy
## B1                      0.5
## B2                     -0.5
```

```
model_y <- lm(DV~ 1 + VariableB, data=final)
summary(model_y)
```

```
##
## Call:
## lm(formula = DV ~ 1 + VariableB, data = final)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.8317 -0.6488  0.0840  0.7083  2.2783
##
## Coefficients:
##                                     Estimate Std. Error t value
## (Intercept)                       -0.00900     0.09077   -0.099
## VariableBNovice versus Expert on Study Strategy -0.82133     0.18154   -4.524
##                                     Pr(>|t|)
## (Intercept)                          0.921
## VariableBNovice versus Expert on Study Strategy 0.0000145 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9943 on 118 degrees of freedom
## Multiple R-squared:  0.1478, Adjusted R-squared:  0.1406
## F-statistic: 20.47 on 1 and 118 DF,  p-value: 0.00001454
```

Tukey Test

```
TukeyHSD(aov(model_1))
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = model_1)
##
## $VariableA
##          diff          lwr          upr          p adj
## A2-A1 0.2426667 -0.100936 0.5862693 0.1645405
##
## $VariableB
##          diff          lwr          upr          p adj
## B2-B1 0.8213333 0.4777307 1.164936 0.0000063
##
## $`VariableA:VariableB`
##          diff          lwr          upr          p adj
## A2:B1-A1:B1 -0.3393333 -0.97885346 0.3001868 0.5124467
## A1:B2-A1:B1 0.2393333 -0.40018679 0.8788535 0.7636396
## A2:B2-A1:B1 1.0640000 0.42447988 1.7035201 0.0001800
## A1:B2-A2:B1 0.5786667 -0.06085346 1.2181868 0.0910313
## A2:B2-A2:B1 1.4033333 0.76381321 2.0428535 0.0000005
## A2:B2-A1:B2 0.8246667 0.18514654 1.4641868 0.0057208
```